

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 21

Level 2 — Intermediate · Number Sense I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Number Sense I — Large Numbers & Place Value
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	21 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Welcome to Level 2! Worksheet 21 steps up to 5-digit numbers and one lakh. The questions now ask for DIFFERENCES and SUMS of place values (not just one place value), building numbers from clues, digit-replacement puzzles, and counting questions like 'how many times do you write the digit 1' — the style where the IMO starts separating rote learners from thinkers.

Learning Objectives

- Work with 5-digit numbers and the lakh in the Indian system.
- Find sums and differences of place values within one number.
- Build numbers from place-value clues and digit conditions.
- Count numbers in ranges and digits in sequences.
- Order large numbers and solve digit-replacement inequalities.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

- Q1.** What is the place value of **6** in **2,63,415** (Indian system)?
(A) 6,000 (B) 60,000
(C) 600 (D) 6,00,000
- Q2.** What is the DIFFERENCE between the place values of the two 5s in **5,254**?
(A) 4,950 (B) 5,050
(C) 4,500 (D) 5,000
- Q3.** The number **seventy-two thousand five hundred eight** in figures is:
(A) 72,580 (B) 72,508
(C) 7,258 (D) 72,058
- Q4.** What is the SUM of the place values of the two 3s in **30,832**?
(A) 33 (B) 30,030
(C) 30,003 (D) 33,000
- Q5.** Which number is 1,000 MORE than 45,999?
(A) 46,999 (B) 45,000
(C) 55,999 (D) 46,099
- Q6.** What is the SUCCESSOR of 99,999?
(A) 99,998 (B) 1,00,000
(C) 10,000 (D) 99,991
- Q7.** Arrange in ASCENDING order: 36,452 · 36,254 · 36,524 · 36,245
(A) 36,245 · 36,254 · 36,452 · 36,524 (B) 36,254 · 36,245 · 36,452 · 36,524
(C) 36,524 · 36,452 · 36,254 · 36,245 (D) 36,245 · 36,452 · 36,254 · 36,524
- Q8.** Using the digits **7, 0, 5, 2, 8** once each, the GREATEST 5-digit number is:
(A) 87,520 (B) 87,502
(C) 85,720 (D) 78,520
- Q9.** Using the digits **7, 0, 5, 2, 8** once each, the SMALLEST 5-digit number is:
(A) 02,578 (B) 20,578
(C) 25,078 (D) 20,587
- Q10.** What is the FACE value of 8 in 84,326 MINUS the face value of 6?
(A) 2 (B) 80,000
(C) 79,994 (D) 8
- Q11.** How many THOUSANDS are there in 67,000?
(A) 6 (B) 7
(C) 67 (D) 670



Q12. In the Indian system, **1 lakh** equals:

- (A) 10,000 (B) 1,00,000
(C) 10,00,000 (D) 1,000

Q13. Which digit must replace ★ so that **4★,567 > 48,567**?

- (A) 7 (B) 8
(C) 9 (D) 6

Q14. The expanded form **40,000 + 3,000 + 50 + 2** is the number:

- (A) 43,052 (B) 43,502
(C) 40,352 (D) 43,520

Q15. What is the PREDECESSOR of 60,100?

- (A) 60,101 (B) 60,099
(C) 60,090 (D) 60,009

Q16. A number has 5 in the thousands place, 2 in the tens place, 8 in the ten-thousands place, 0 in the hundreds place and 6 in the ones place. The number is:

- (A) 85,026 (B) 58,026
(C) 85,206 (D) 82,056

Q17. How many 4-digit numbers are there in all (from 1,000 to 9,999)?

- (A) 8,999 (B) 9,000
(C) 9,999 (D) 8,000

Q18. Which is the GREATEST: $9,999 + 1 \cdot 10,000 - 10 \cdot 9,990 + 5 \cdot 10,001 - 2$?

- (A) $9,999 + 1$ (B) $10,000 - 10$
(C) $9,990 + 5$ (D) $10,001 - 2$

Q19. If you write the numbers from 1 to 20, how many times do you write the digit **1**?

- (A) 10 (B) 11
(C) 12 (D) 13

Q20. A car's odometer shows **45,978 km**. After how many more kilometres will it show 46,000 km?

- (A) 12 (B) 22
(C) 32 (D) 122

ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	A	B	B	A	B	A	A	B	A
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	B	C	A	B	A	B	A	C	B

Step-by-Step Solutions

Q1. Correct answer: **(B)**

In 2,63,415 the 6 is in the ten-thousands place: $6 \times 10,000 = \mathbf{60,000}$.

Q2. Correct answer: **(A)**

First 5 (thousands) = 5,000; second 5 (tens) = 50. Difference = $5,000 - 50 = \mathbf{4,950}$.

Q3. Correct answer: **(B)**

72 thousand = 72,000; five hundred eight = 508. Total **72,508** — note the 0 in the tens place.

Q4. Correct answer: **(B)**

First 3 (ten-thousands) = 30,000; second 3 (tens) = 30. Sum = **30,030**.

Q5. Correct answer: **(A)**

$45,999 + 1,000 = \mathbf{46,999}$ (only the thousands digit changes: 5 → 6).

Q6. Correct answer: **(B)**

$99,999 + 1 = \mathbf{1,00,000}$ (one lakh) — all the 9s roll over.

Q7. Correct answer: **(A)**

All start 36,—. Compare hundreds: 2,2,4,5 → the two '36,2..' first. Then tens: $36,245 < 36,254$. Order: **36,245 · 36,254 · 36,452 · 36,524**.

Q8. Correct answer: **(A)**

Arrange from biggest: 8, 7, 5, 2, 0 → **87,520**.

Q9. Correct answer: **(B)**

A number cannot start with 0. Smallest non-zero digit first (2), then 0, then the rest ascending: **20,578**.

Q10. Correct answer: **(A)**

Face values are just the digits: $8 - 6 = \mathbf{2}$. (Face value ignores position.)

Q11. Correct answer: **(C)**

$67,000 = 67 \times 1,000$, so there are **67 thousands**.

Q12. Correct answer: **(B)**

1 lakh = **1,00,000** (one hundred thousand).

Q13. Correct answer: **(C)**

The number is 4★,567 vs 48,567. For it to be GREATER, ★ must be more than 8 → ★ = **9** ($49,567 > 48,567$).

Q14. Correct answer: **(A)**

$40,000 + 3,000 = 43,000$; $+ 50 + 2 = \mathbf{43,052}$. The hundreds place gets a 0.



Q15. Correct answer: (B)

$60,100 - 1 = 60,099$.

Q16. Correct answer: (A)

Place the digits: ten-thousands 8, thousands 5, hundreds 0, tens 2, ones 6 → **85,026**.

Q17. Correct answer: (B)

Count = $9,999 - 1,000 + 1 = 9,000$ numbers.

Q18. Correct answer: (A)

Compute: 10,000; 9,990; 9,995; 9,999. The greatest is **$9,999 + 1 = 10,000$** .

Q19. Correct answer: (C)

1, 10, 11 (twice!), 12, 13, 14, 15, 16, 17, 18, 19 → that is $1 + 1 + 2 + 8 = 12$ times.

Q20. Correct answer: (B)

$46,000 - 45,978 = 22$ km.

